

The shadow-screen lab

Teacher guide and worked answers

Explain that a shadow forms where an object blocks light from a source.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsb/inputs/Build SOW 2026-2027.xlsx | BUILD Weekly - Spring | C36

Investigate how shadows are formed when light is blocked.

Direct match to the named outcome. The lesson is a 40-minute exemplar, with optional greater depth; no accreditation completion is claimed.

Prior learning

Everyday objects can be described by what we observe.

Use a labelled diagram or evidence card to explain a choice.

Success evidence

Place a blocker between source and screen.

Explain a shadow using blocked light.

Use evidence to compare materials without overclaiming.

Equipment

Low-power LED torch; white screen; card rectangle; clear plastic sample; translucent paper; stands; printed cases.

Preparation

Pre-cut equal-sized samples with safe edges. Fix source, sample and screen positions.

Check that the screen shows a useful contrast with room lights comfortably on; supply the evidence cards if not.

Practical care

Low-power cool LED only, aimed at the screen below face level. No Sun, lasers or hot lamps.

Keep floor routes lit. No pupil works in complete darkness.

Use safe plastic, not glass. Do not place materials over a hot light.

Ready to teach alternative

Use the constructed observations and ray diagrams. Actual visibility varies with materials, lighting and source size.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Build a screen silhouette

Slide	Stage	Minutes	Focus
2	Retrieval	2-5	Three clues for our design
3	I do	5-9	I start without a blocker
4	I do	9-12	I put a card in the path
5	I do	Within stage	Materials let different amounts through
6	We do	12-16	Choose a silhouette material
7	We do	16-20	The disappearing-shape challenge
8	Hinge	20-23	Why is there a shadow?
9	You do	23-25	Choose your investigation route
10	You do	Within stage	Use the evidence
11	You do	25-35	Make your own science record
12	You do	Within stage	Check and improve
13	Review	35-38	Lundy loop: improve our account
14	Exit	38-40	Three quick checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Build a screen silhouette

Pose a stage-screen design brief. Pupils will choose a sample for a clear silhouette and explain how it works. No performance is required.

2 Retrieval Three clues for our design

Accept a valid source; reflection returns light from a surface; an opaque card blocks transmission. Explain that reflecting and blocking transmission can occur together.

3 I do I start without a blocker

Think aloud: "I need a comparison. Before adding the card, I check that the source lights the screen. If it does not, I repair the setup before judging a material."

4 I do I put a card in the path

Think aloud: "I trace the light towards the card. Some paths are stopped; rays past its edges still reach the screen. The shadow is on the screen, not a dark substance leaving the card." Other room light can make it grey rather than black.

5 I do Materials let different amounts through

Shares I do time. Think aloud: "I judge the observation in this setup. Clear plastic may still give a faint edge or pattern; I will not claim it can never affect light." These arrows are qualitative, not counts of photons or measured percentages.

6 We do Choose a silhouette material

Constructed evidence comes from the same source, screen and sample position. A opaque card is the best supported choice for the stated goal. Use reader, chooser and reason-giver roles; no competition.

Reveal: A: the opaque card gave the clearest dark shape in these cases.

7 We do The disappearing-shape challenge

Let pupils move a card on a tabletop or use the case. A shadow from this card disappears from the tested screen area when it no longer blocks that path. Ask why a black and a red opaque card of equal shape can give similar silhouettes.

Reveal: Removing the blocker lets light reach that area again.

8 Hinge Why is there a shadow?

If pupils choose dark paint, remove and return the card in the model. If source goes off, show the ray paths still reaching around the edges.

A. The card sends dark paint onto the screen. - No material travels from the card; some light paths are blocked.

B. The card blocks light, so less reaches part of the screen. - This is the shadow region; light can still reach around the card.

C. The source switches off when the card arrives. - The source can stay on. The card blocks only some paths.

9 You do Choose your investigation route

Give one route only, selected with the pupil. The shared evidence page supports every route. Optional practical: test pre-cut samples at the fixed position, leaving the source and screen still. Record actual observations rather than copying sample results. If contrast is poor, use the supplied cases and discuss the limitation.

10 You do Use the evidence

Shares the 12-minute independent stage. Optional practical: test pre-cut samples at the fixed position, leaving the source and screen still. Record actual observations rather than copying sample results. If contrast is poor, use the supplied cases and discuss the limitation.

11 You do Make your own science record

Allow ten minutes of independent response after the two-minute setup. Optional practical: test pre-cut samples at the fixed position, leaving the source and screen still. Record actual observations rather than copying sample results. If contrast is poor, use the supplied cases and discuss the limitation.

12 You do Check and improve

Shares the independent stage. Ask the pupil to point to the clue supporting their answer; accept a spoken correction or changed choice.

13 Review Lundy loop: improve our account

Give a private or paired option. Adopt one scientifically sound pupil revision in the class explanation and name the change. No public ranking.

Reveal: It is not a substance: it is a region where less light reaches because light is blocked.

14 Exit Three quick checks

Collect brief individual answers. Secure core requires the main idea and a relevant reason. Revisit the relevant worked diagram next lesson if the explanation reverses cause and effect.

Reveal: See the teacher answer guide for the three checks.

Answers and assessment

Retrieval 1-3

1 valid light source. 2 returns light from a surface. 3 yes, an opaque card stops light passing through.

We do W1-W2

W1 A; it has the clear dark shape in the evidence. W2 light reaches the tested screen area again, so that card's shadow there disappears.

Hinge

Second option correct. The card blocks some light paths; it does not send paint or switch off the source.

S1-S5

S1 A. S2 blocks. S3 source → card → screen. S4 on the screen. S5 light reaches the area again; that shadow disappears.

T1-T3

T1 A blocks transmission and makes dark shape; B transmits some and scatters it; C transmits much light but can show a faint edge. T2 correct relative positions, blocking at the card, shadow behind it on screen. T3 the supplied clear sample has a faint visible edge; clear materials can redirect or reflect some light.

X1-X4

X1 the opaque card blocks light, reducing light reaching part of the screen. X2 similar dark silhouette in this simple setup; the card's red colour is not projected as a red picture. X3 room light or other reflected light can still reach the shadow area. X4 move card out and back while source stays on, or compare equal opaque shapes of different colours.

Exit E1-E3

E1 blocks it from passing through. E2 on the screen behind the blocker relative to source. E3 light reaches that area again; the shadow from that blocker is removed.

R1-R2

Responses vary. Credit a science-related revision or relevant next question. A pass or private response is acceptable; preferences have no fixed right answer.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 7; We do 8; hinge 3; You do 12; review 3; exit 2. Zero-minute slides share the stage time.

BUILD follows the user's Year 3 science anchor with age-respectful materials for secondary pupils. Routes are adjustable access choices, not fixed groups or diagnoses.

Every core task can run from the printed evidence. Do not present constructed teaching evidence as observations pupils made. A pupil's spoken reasoning carries the same scientific value as writing.

Do not force the wording "no shadow" for a transparent sample. Clear materials reflect and refract some light; surface edges and imperfect transmission may create faint patterns.

Success rests on a causal explanation, not remembering the longest vocabulary word. Accept "does not let the light through" for opaque.

HTML We do: pupils choose a prediction, test it against the user-controlled SVG model or supplied evidence, then explain or revise. The shared paper cases and W1/W2 prompts provide the equivalent

classroom discussion without a device. Any animation is a teaching model, not filmed experimental evidence.

Access and responsive teaching

Supported

Choose one route: short choices, word bank and pointing or adult-scribed explanation; keep the same science goal.

Standard

Use the evidence provided to record a decision and explain the science in short sentences.

Stretch

Apply the same core idea to a changed setup, identify a limitation and justify a better decision.

Regulation

Preview the sequence and equipment. Offer seated observation, a quiet paper route, partner rehearsal and a pass. Never require public performance or a darkened room.

Misconceptions to check

A shadow is a black object sent out by the card.

A shadow is a region receiving less light because light is blocked.

Check: Remove the card: what changes on the screen?

Transparent materials can never cast a shadow.

They can cause faint shadows or patterns; much light passes through clear material.

Check: Use the cautious wording on card C.

The shadow always has the object's colours.

An opaque object blocks light; its surface colours are not projected by this simple setup.

Check: Compare two equal opaque shapes with different colours.

Word	Meaning
shadow	A darker area where less light reaches a surface because light is blocked.
opaque	Does not let light pass through.
translucent	Lets some light through, scattering it.
transparent	Lets much light through so objects can be seen clearly through it.

Scientific references

Department for Education: [science programmes of study](#)

Checked 8 September 2026. Year 3 light outcomes anchor the BUILD content; ray explanations are carefully modelled as an accessible extension.